Seat No.:________ Enrolment No.__________

GUJARAT TECHNOLOGICAL UNIVERSITY

BE - SEMESTER– III (New) EXAMINATION – WINTER 2019

Subject Code: 3130608 Date: 5/12/2019

Subject Name: Mechanics of Solids

Time: 02:30 PM TO 05:00 PM Total Marks: 70

Instructions:

1. **Attempt all questions.**
2. **Make suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**

			MARKS
Q.1*	(a)	Define: (1) Rigid body, (2) Newton's second Law	03
	(b)	Define Force and classify the force system with neat sketch.	04
	(c)	Find magnitude and direction of resultant of force system shown in Fig.1	07
Q.2	(a)	Define Moment & Couple giving two suitable examples	03
	(b)	State Hook's low. Draw stress strain curve for Mild Steel Specimen and explain each point in detail.	04
	(c)	A Chord supported at A,B carries a load of 20kN at point C and unknown weight W kN at D as shown in fig 2.Find the value of unknown weight W. So that CD remains horizontal.	07
		OR	
	(c)	Determine Ixx and Iyy for section shown in fig 3	07
Q.3	(a)	Define (i) Strain (ii) Poisson's ratio (iii) Bulk Modulus	03
	(b)	Find support reaction for a beam as shown in figure. 4	04
	(c)	A Reinforced concrete column is applied 700 kN load. Size of column is 300 mm X 400 mm, and it is reinforced with 6 bars of 16 mm dia. Determine load taken by concrete and steel.	07
		OR	
Q.3	(a)	Define (1) Ductile material (2) Compound bar (3) Axial load	03
	(b)	Find support reactions for a beam as shown in figure. 5	04
	(c)	A 2.8 m long member is 60 mm deep and 40 mm wide. It is subjected to axial tensile force 210 kN. Determine change in dimension and in volume. Take E=200 Gpa and $\mu = 0.3$ Assume Esteel and Econcrete	07
Q.4	(a)	Derive the formula for the elongation of a rectangular bar under the action of axial load.	03
	(b)	Explain with neat sketch types of beams, types of loads and types of supports	04
	(c)	A steel rod 25mm in diameter is inserted inside a brass tube of 25mm internal diameter and 35mm external diameter, the ends are rigidly connected together. The assembly is heated by 30^0C. Find value and nature of stress developed in both the materials. Take, E_{steel} = 200GPa, E_{brass} =80 GPa, α_{steel} =12 x 10-6 per ^{0}C, α brass =18 x 10-6 per ^{0}C.	07
		OR	
Q.4	(a)	Write the assumption made in theory of pure torsion.	03
	(b)	Derive the equation for deformation of a body due to self weight.	04
	(c)	Draw Shear Force and Bending Moment diagram for the beam shown in fig. 6	07
Q.5	(a)	Define: 1)Point of Contra flexure, 2) Shear force	03
	(b)	Derive assumption made in analysis of truss.	04
	(c)	Analysis the truss loaded as shown in figure. 7	07

OR

Q.5	(a)	\V*n te* a differ *ence* between *Truss* and frame.	03
	(b)	Explain perfect truss and imperfect truss with the sketches.	04
	(c)	Analysis the truss loaded as shown in figure 8	07

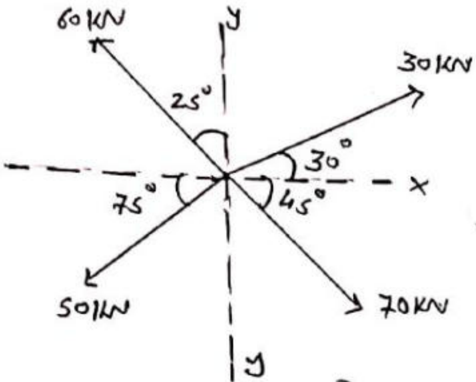

Fig. 1 Q.1 (c)

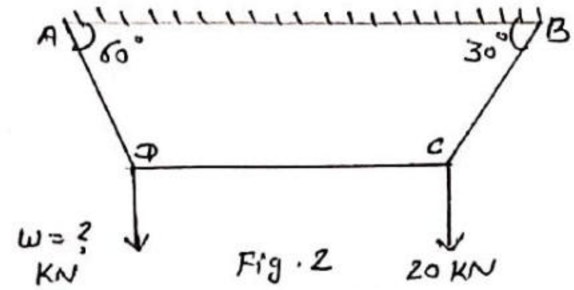

Fig. 2
Q.2 (c)

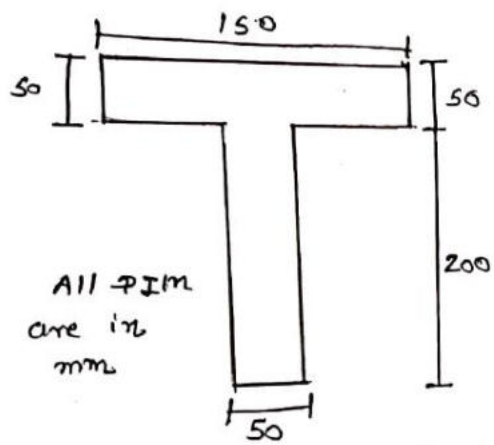

Fig. 3 Q.2 (c)

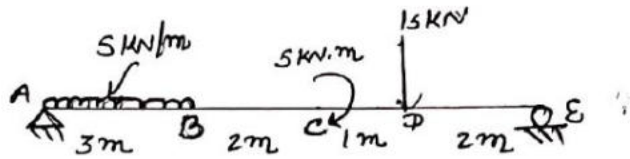

Fig. 4 Q.3 (b)

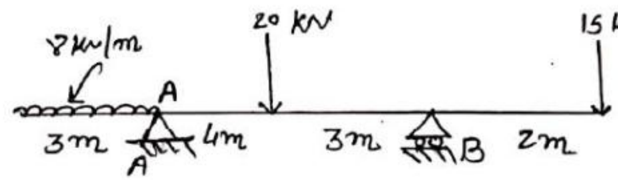

Fig. 5 Q.3 (b)

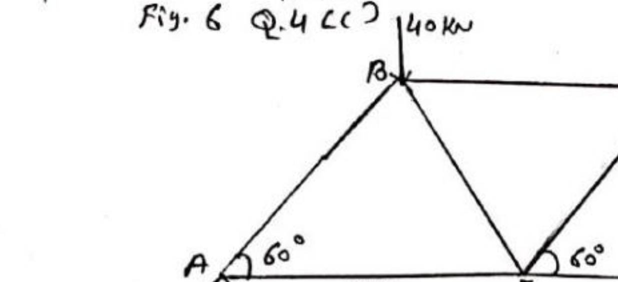

Fig. 6 Q.4 (c)

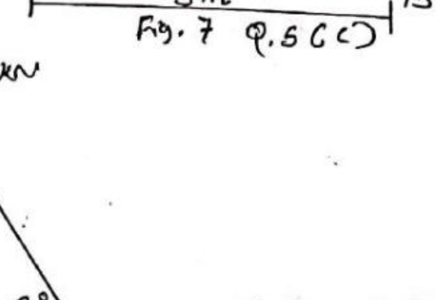

Fig. 7 Q.5 (c)

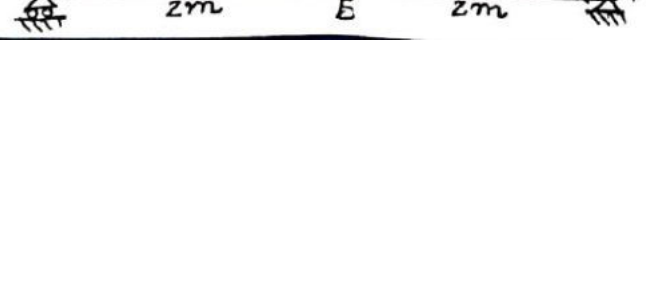

Fig. 8 Q.5 (c)

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GUJARAT TECHNOLOGICAL UNIVERSITY

BE - SEMESTER– III EXAMINATION – SUMMER 2020

Subject Code: 3130608 **Date:04/11/2020**

Subject Name: Mechanics of Solids

Time: 02:30 PM TO 05:00 PM **Total Marks: 70**

Instructions:

1. **Attempt all questions.**
2. **Make suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**

			Marks
Q.1	**(a)**	State's Law of Parallelogram of forces.	**03**
	(b)	Define force and writes its characteristics.	**04**
	(c)	Find the magnitude and direction of resultant of force system shown in fig. 01.	**07**
Q.2	**(a)**	What is meant by free body diagram? Draw free body diagram for box place on a table.	**03**
	(b)	Define : (1) Isotropic material (2) Anisotropic material (3) Homogeneous material (4) Orthotropic material.	**04**
	(c)	Find the minimum (least) value of force P to keep the sphere in the position shown in fig. 02. The radius of sphere 1 is 5cm and sphere 2 is 10cm. The weight of sphere 1 is 100N and sphere 2 is 200N.	**07**
		OR	
	(c)	Draw shear force diagram and bending moment diagram for a beam shown in fig. 03.	**07**
Q.3	**(a)**	What is difference between deficient truss and redundant truss.	**03**
	(b)	Explain types of supports with usual notations.	**04**
	(c)	Find the CG of plane lamina shown in fig 4.	**07**
		OR	
Q.3	**(a)**	Explain : (1) Poisson's ratio (2) Hook's law (3) Bulk modulus.	**03**
	(b)	A bar of 3m long and 20mm diameter is rigidly fixed in two supports at certain temperature. If temperature is raised by 60° C, find the thermal stress and strain of the bar. Also find thermal stress and strain if support yields by 2 mm. Take $\alpha = 12 \times 10^{-6}$ / °C and $E = 2 \times 10^5$ N/mm^2.	**04**
	(c)	State and explain with figure Pappu's –Guildinus theorem of surface area of Revolution.	**07**
Q.4	**(a)**	Enlist the assumptions made in theory of torsion.	**03**
	(b)	A beam simple supported and carries an U.D.L. of 50 kN/m over whole span. The size of beam is 150mm x 400mm.. If maximum stress in the material of beam is 100N/mm^2 find the span of beam.	**04**
	(c)	Determine the centroid of the section shown in fig. 05.	**07**

OR

Q.4	**(a)**	A load of 10 kN is to be raised with help of a steel wire. Find the minimum diameter of the wire, if the stress is not to be exceed 80 N/mm^2.	**03**
	(b)	Explain types of beams with notations.	**04**
	(c)	Determine moments of inertia of a section shown in fig. 06 about horizontal centroidal axis.	**07**
Q.5	**(a)**	Define: (1) Shear Force (2) Bending Moment (3) Points of contraflexure	**03**
	(b)	Derive the relation between : (1) Young's Modulus (2) Modulus of Rigidity (3) Possion's Ratio	**04**
	(c)	A hollow steel shaft, 3m of length must transmit a torque of 25 kNm. The total angle of twist in this length is not to exceed 2.5° and the allowable shearing stress in the material is 90 MPa. Calculate the inside diameter of the shaft and thickness of the metal. G = 85 GN/m^2.	**07**

OR

Q.5	**(a)**	Draw shear stress distribution diagram for : (1) I section (2) Circular section (3) Triangular section	**03**
	(b)	Explain assumptions made in theory of pure bending.	**04**
	(c)	A square prism of metal 60mm x 60mm in cross section and 300mm long is subjected to a tensile stress of 450 MPa along its longitudinal axis, lateral compression of 240 MPa and lateral tension of 120 MPa along the pair of sides. If E = 150 GPa, calculate the changes in dimensions, change in volume of metal. μ = 0.36.	**07**

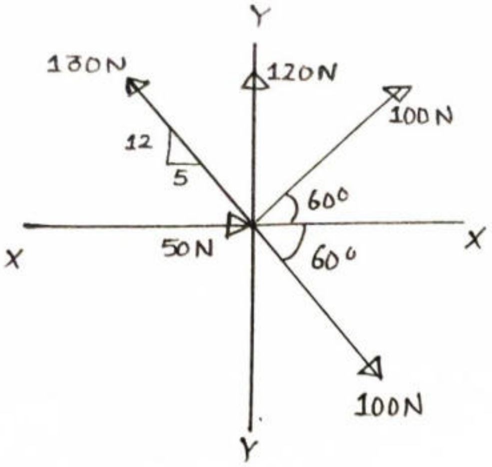

Fig. 01

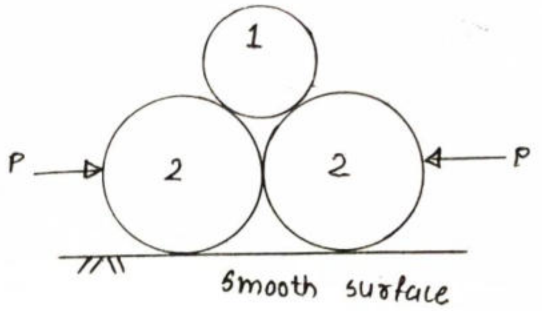

Fig. 02

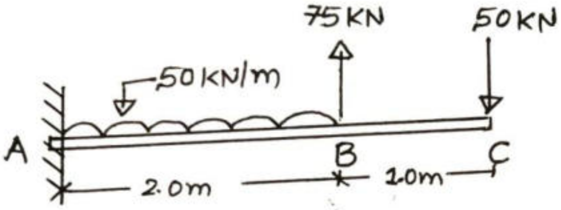

Fig. 03

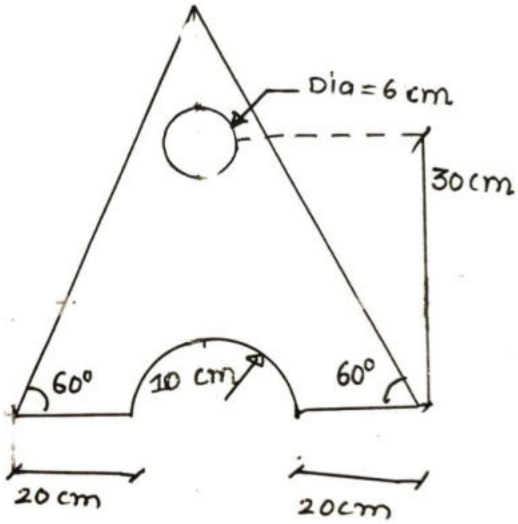

Fig. 04

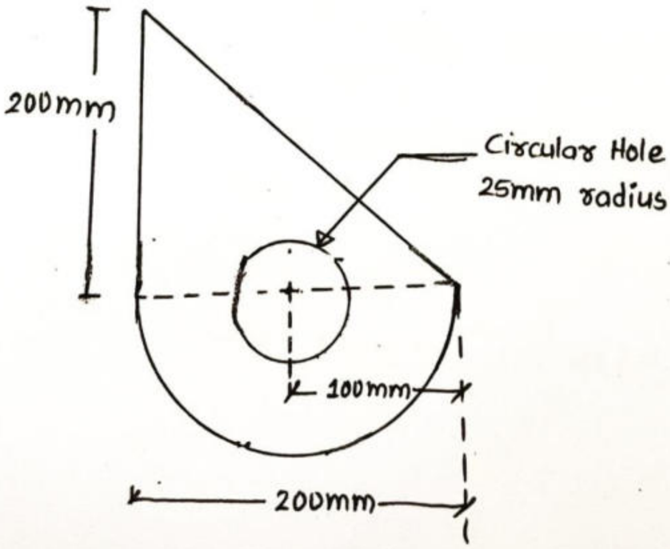

Fig. 05

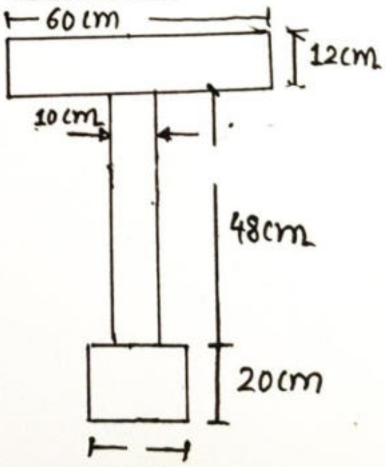

Fig. 06

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GUJARAT TECHNOLOGICAL UNIVERSITY

BE- SEMESTER–III (NEW) EXAMINATION – WINTER 2020

Subject Code:3130608 Date:06/03/2021

Subject Name:Mechanics of Solids

Time:10:30 AM TO 12:30 PM Total Marks:56

Instructions:

1. **Attempt any FOUR questions out of EIGHT questions.**
2. **Make suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**

Q.1	**(a)**	Explain following terms (i) Rigid body, (ii) Deformable body (iii) Elastic body.	**03**
	(b)	State Hook's low. Draw stress strain curve for Mild Steel Specimen and explain each point in detail.	**04**
	(c)	Three forces are acting on a weightless equilateral triangular plate as shown in **Fig. 1.** Determine magnitude, direction and position of resultant force.	**07**
Q.2	**(a)**	Explain : (i) Type of beams (ii) Type of loading on the beams.	**03**
	(b)	Determine support reaction for the given beam shown in **Fig. 2.**	**04**
	(c)	A simply supported beam 10 m long carries three point loads at 100 kN, 150 kN and 200 kN at 3m, 5m and 8m from left support. Draw S.F. and B.M. diagram for the beam.	**07**
Q.3	**(a)**	Discuss critically the assumption made in theory of Bending.	**03**
	(b)	A cantilever beam 2 m long has rectangular section 200 mm x 500 mm. Find out point load at free end of beam if permissible bending stress is $20 N/mm^2$.	**04**
	(c)	A beam of I-section, having 5 m length is simply supported at each end and bears a u.d.l. of 20 kN/m as shown in **Fig. 3**. Determine maximum tensile and compressive bending stress.	**07**
Q.4	**(a)**	Derive with usual notations the theorem of perpendicular axis.	**03**
	(b)	Derive relation between bulk modulus (K), Poisson's ratio (1/m), and modulus of elasticity (E).	**04**
	(c)	A beam of I-section, having 5 m length is simply supported at each end and bears a u.d.l. of 20 kN/m as shown in **Fig. 3.** Determine maximum shear stress.	**07**
Q.5	**(a)**	Define: (1) Centroid (2) Center of gravity (3) Center of mass.	**03**
	(b)	State and prove Pappu's guldinus theorem for surface area of bodies.	**04**
	(c)	Determine the location of centroid of plane lamina shown in **Fig. 4** with respect to point O.	**07**
Q.6	**(a)**	Write assumption made in the theory of torsion.	**03**
	(b)	A solid steel shaft is to transmit 120 kW power at 600 r.p.m. Find the diameter of shaft if shear stress is to be limited to $100 N/mm^2$.	**04**
	(c)	Determine moment of inertia about base of a plane area as shown in **Fig. 4.**	**07**

Q.7	**(a)**	Define : (i) Modulus of Elasticity (ii) Poisson's ratio(iii) Modulus of rigidity	**03**
	(b)	In a tension test, a bar of 20 mm diameter undergoes elongation of 14 mm in a gauge length of 150 mm and a decrease in diameter of 0.85 mm at a tensile load of 6 kN. Determine the two physical constants Poisson's ratio and modulus of elasticity of the material.	**04**
	(c)	A steel rod 30 mm in diameter is inserted inside a brass tube of 30 mm internal diameter and 40 mm external diameter, the ends are rigidly connected together. The assembly is heated by 20°C. Find value and nature of stress developed in both the materials. Take, α for steel = 12 x 10^{-6} per °C, α for brass = 18 x 10^{-6} per °C. E for steel = 200 GPa, E for brass = 80 GPa,	**07**
Q.8	**(a)**	Define principal planes and principal stresses.	**03**
	(b)	A R.C.C. column 300 mm in dia. is reinforced with 6 nos. of 16 mm diameter steel bars. If permissible stress in steel and concrete are 230 N/mm^2 and 5 N/mm^2, respectively, find the load carrying capacity of the column.	**04**
	(c)	The state of stress in two-dimensionally stress body at a point is shown in **Fig. 5.** Determine principal stresses and maximum shear stress and its location of planes.	**07**

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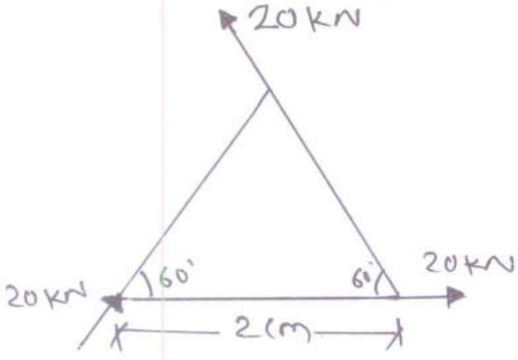

Fig-1.

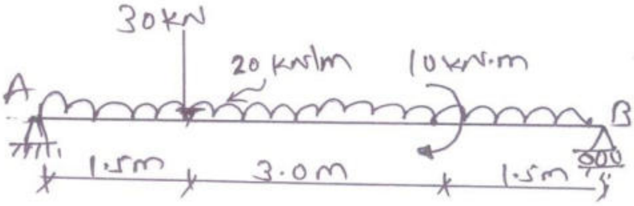

Fig-2.

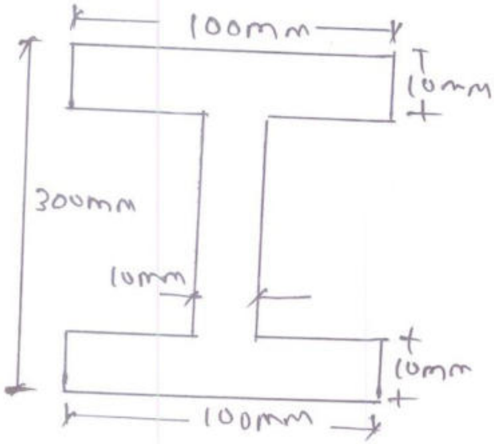

Fig-3

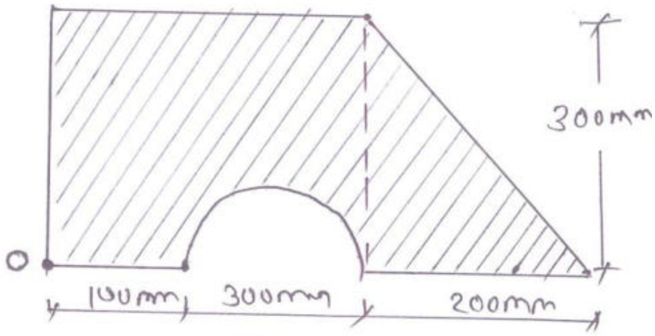

Fig-4

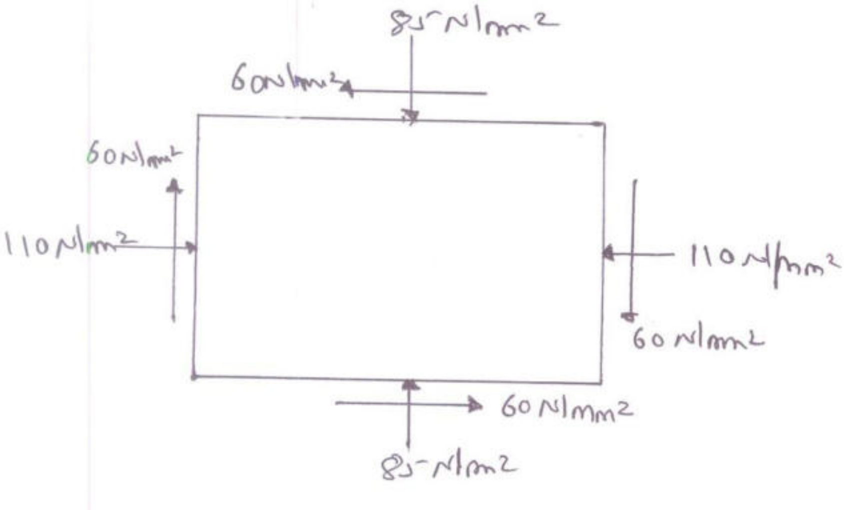

Fig. 5

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GUJARAT TECHNOLOGICAL UNIVERSITY

BE - SEMESTER–III (NEW) EXAMINATION – SUMMER 2021

Subject Code:3130608 **Date:16/09/2021**

Subject Name:Mechanics of Solids

Time:10:30 AM TO 01:00 PM **Total Marks:70**

Instructions:

1. **Attempt all questions.**
2. **Make suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**
4. **Simple and non-programmable scientific calculators are allowed.**

			MARKS
Q.1	**(a)**	State the characteristics of couple.	**03**
	(b)	State and explain parallelogram law of forces.	**04**
	(c)	Calculate magnitude and direction of the resultant force of the force system shown in **Figure-1**. Each square is of size 10 cm × 10 cm as shown in Figure-1.	**07**

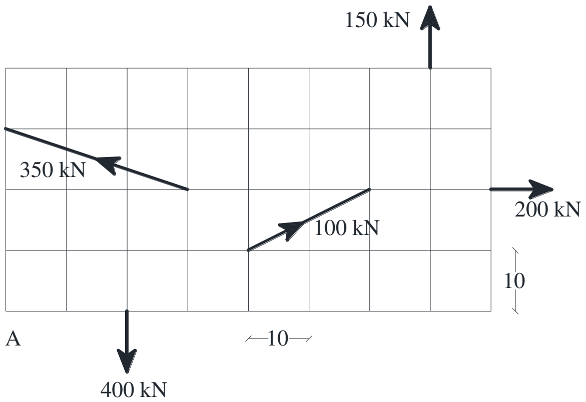

Figure-1

Q.2	**(a)**	State and explain Lami's theorem.	**03**
	(b)	Calculate the point of application of resultant force of the force system shown in **Figure-1**. Each square is of size 10 cm × 10 cm as shown in Figure-1.	**04**
	(c)	Determine the support reactions for the beam shown in **Figure-2**. Also plot the shear force and bending moment diagrams.	**07**

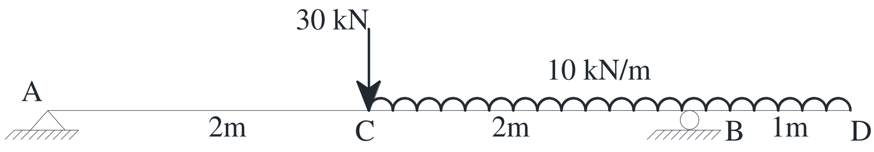

Figure-2

OR

	(c)	Determine the support reactions for the beam shown in **Figure-3**. Also plot the shear force and bending moment diagrams.	**07**

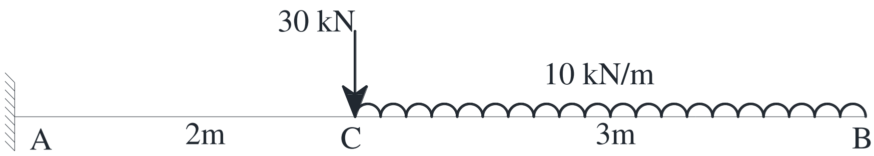

Figure-3

Q.3	**(a)**	Mention the assumption made in the theory of pure bending.	**03**

(b) Derive using first principle the equation for calculation of maximum shear stress at a section for a beam with rectangular cross section. **04**

(c) Calculate the maximum bending stress and maximum shear stress at a section for the beam shown in **Figure-3**. Beam cross section is 300mm × 500mm deep. **07**

OR

Q.3 (a) Write the assumptions made in the analysis of perfect truss. **03**

(b) State and explain Verignon's principle. **04**

(c) Calculate the maximum bending stress and maximum shear stress at a section for the beam shown in **Figure-2**. Beam cross section is 300mm wide × 500mm deep. **07**

Q.4 (a) State parallel axes and perpendicular axes theorems. **03**

(b) Derive torsion equation with usual notations. **04**

(c) Locate the centroid from the given reference axes of the lamina shown in **Figure-4**. **07**

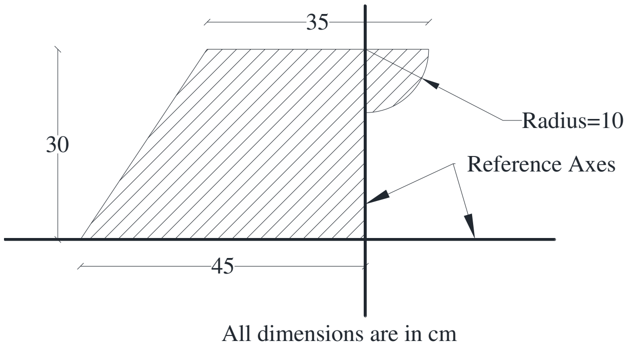

Figure-4

OR

Q.4 (a) State assumptions made in theory of torsion. **03**

(b) Derive the equation to locate centroid of the semicircular lamina. **04**

(c) Calculate the moment of inertia of the section about the horizontal axis passing through the base. Refer **Figure-4**. **07**

Q.5 (a) State and explain Hook's law. **03**

(b) A brass rod ABCD having a c/s area of 500mm^2 is subjected to axial forces as shown in **Figure-5**. If modulus of elasticity for brass is 80kN/mm^{2,} find the change in the length of the bar. **04**

Figure-5

(c) The tensile stresses at a point across two mutually perpendicular planes are 120N/mm^2 and 60N/mm^2. Determine the normal, tangential and resultant stresses on a plane inclined at 30° to the axis of the minor stress. **07**

OR

Q.5 (a) Describe the Mohr's circle method to calculate principal stresses. **03**

(b) A brass rod ABCD having a c/s area of 500mm^2 is subjected to axial forces as shown in **Figure-5**. If modulus of elasticity for brass is 80kN/mm^{2,} find stress in each part of the bar. **04**

(c) A rectangular block of material is subjected to tensile stresses of 110MPa and 50MPa. Each of these stresses is accomplished by shear stress of 60MPa which tends to rotate block in anticlockwise direction. Determine direction and magnitude of Major and Minor principal stress on oblique plane. **07**

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Seat No.: ________ Enrolment No.___________

GUJARAT TECHNOLOGICAL UNIVERSITY

BE - SEMESTER–III (NEW) EXAMINATION – WINTER 2021

Subject Code:3130608 Date:25-02-2022

Subject Name:Mechanics of Solids

Time:10:30 AM TO 01:00 PM Total Marks:70

Instructions:

1. **Attempt all questions.**
2. **Make suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**
4. **Simple and non-programmable scientific calculators are allowed.**

Q.1	**(a)**	State and explain Lami's theorem.	**03**
	(b)	Determine resultant of given force system shown in **Fig. 1.**	**04**
	(c)	The forces are acting on a rigid body as shown in **Fig. 2.** Find the resultant of the given force system, in terms of magnitude and direction. Find perpendicular distance of resultant force with respect to point A.	**07**
Q.2	**(a)**	Explain various types of statically determinant beams and their support system.	**03**
	(b)	Differentiate between joint method and section method in analysis of plane truss.	**04**
	(c)	Draw Shear Force and Bending Moment diagram for the beam shown in **Fig. 3.**	**07**
		OR	
	(c)	Draw Shear Force and Bending Moment diagram for the beam shown in **Fig. 4.**	**07**
Q.3	**(a)**	Draw shear stress distribution diagram for rectangular, circular and I section.	**03**
	(b)	A beam simply supported over a span of 8 m and carries an U.D.L. of 50 kN/m over whole span. The size of beam is 150 mm x 400 mm. Find the maximum bending stress and draw the bending stress diagram.	**04**
	(c)	A beam of T shaped cross section shown in **Fig. 5** is subjected to bending about x-x axis due to a moment of 20 kN.m. Find the bending stress at top and bottom of the beam.	**07**
		OR	
Q.3	**(a)**	Write assumptions made in theory of pure bending.	**03**
	(b)	Find the support reaction for beam which is loaded as shown in **Fig. 6.**	**04**
	(c)	A beam of T shaped cross section shown in **Fig. 5** is subjected to maximum shear force of 50 kN. Determine maximum shear stress.	**07**
Q.4	**(a)**	Derive with usual notations the theorem of perpendicular axis.	**03**
	(b)	Write equation of Moment of inertia for rectangular section and triangular section about its neutral axis and base of section.	**04**
	(c)	Determine the location of centroid of wire which is bent as shown in **Fig. 7.**	**07**
		OR	
Q.4	**(a)**	Write assumption made in the theory of torsion.	**03**
	(b)	External and internal diameter of propeller shaft are 400 mm and 200 mm respectively. Find maximum shear stress developed in the cross section when a twisting moment of 50 kN.m is applied.	**04**
	(c)	Determine moment of inertia about (I_{xx}) of a plane area as shown in **Fig. 8.**	**07**

Q.5 **(a)** Define (i) Stress (ii) Strain (iii) Modulus of elasticity. **03**

(b) An underground cable is laid in a summer at 32° C. What stress would be induced in it when temperature in winter is -3° C. The cable is unable to contract in any direction. Take, $\alpha = 12 \times 10^{-6}$ per °C, El = 200 GPa. **04**

(c) A steel rod of 100 mm in diameter is inserted inside a copper tube of 200 mm external diameter and 100 mm internal diameter. The composite section is subjected to axial tensile force of 100 kN. Length of the composite section is 0.5 m. Calculate stress in each material. Take E for steel = 2.1×10^5 MPa, E for copper = 1.3×10^5 MPa. **07**

OR

Q.5 **(a)** Define principal planes and principal stresses. **03**

(b) A steel bar 16 mm diameter and 3 m long is subjected to an axial pull of 80 kN. Determine the change in dimension and change in volume of the bar. Take E = = 2×10^5 MPa and $\mu = 0.3$. **04**

(c) The state of stress in two-dimensionally stress body at a point is shown in **Fig. 9.** Determine (i) principal stresses (ii) maximum shear stress and its location of planes. **07**

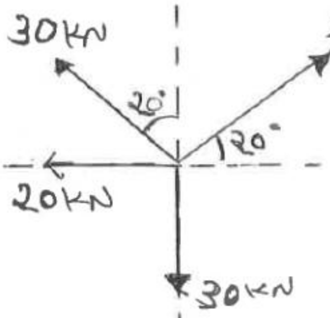

Fig-1

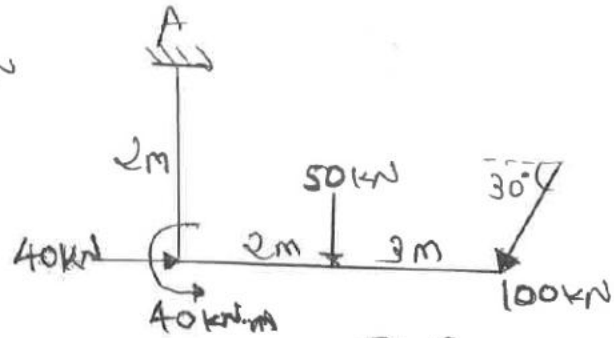

Fig-2

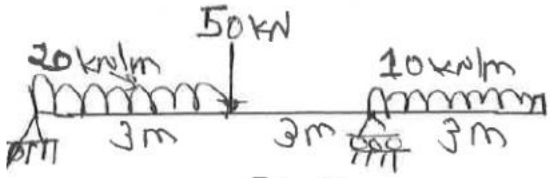

Fig-3

Fig-4

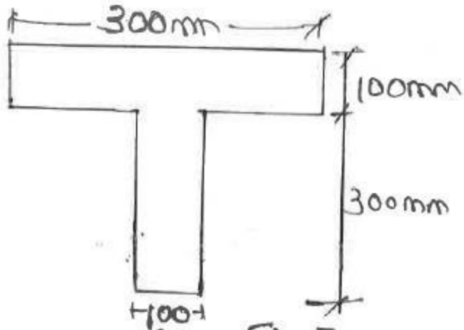

Fig-5

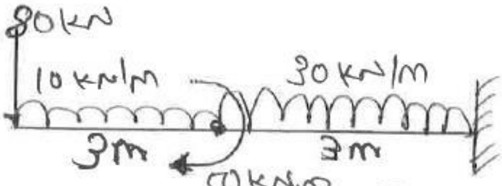

Fig-6

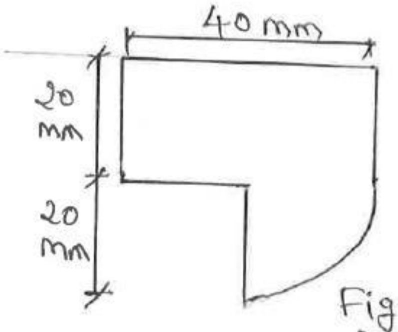

Fig-8

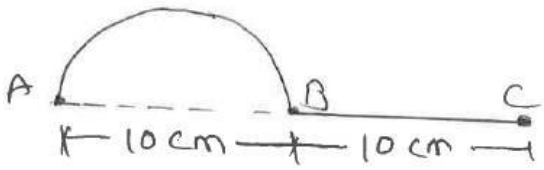

Fig-7

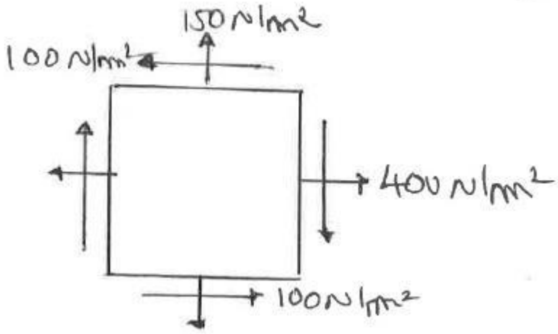

Fig. 9

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Seat No.: ________ Enrolment No.___________

GUJARAT TECHNOLOGICAL UNIVERSITY

BE - SEMESTER– III (NEW) EXAMINATION – SUMMER 2022

Subject Code:3130608 Date:20-07-2022

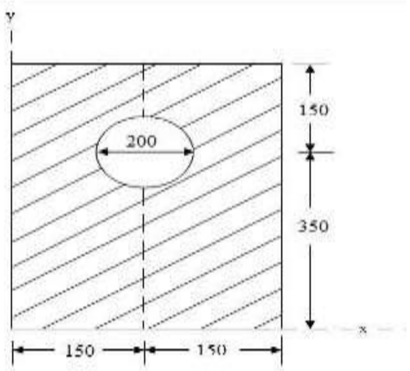
Subject Name:Mechanics of Solids

Time:02:30 PM TO 05:00 PM Total Marks:70

Instructions:

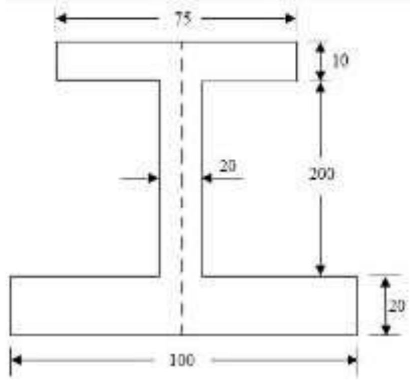
1. **Attempt all questions.**
2. **Make suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**
4. **Simple and non-programmable scientific calculators are allowed.**

			Marks
Q.1	**(a)**	Define: (1) Rigid Body (2) Newton's second law (3) Law of Transmissibility.	**03**
	(b)	State and explain parallelogram law of forces.	**04**
	(c)	The following forces act at a point: (1) 20 N inclined at 30° towards North of East, (2) 25 N towards North, (3) 30 N towards North West, (4) 35 N inclined at 40° towards South of West. Find magnitude and direction of the resultant force.	**07**
Q.2	**(a)**	Differentiate between Moment and Couple.	**03**
	(b)	State and explain Lami's theorem.	**04**
	(c)	Find the moment of inertia of a plate with a circular hole about its centroidal x axis as shown in figure below.	**07**

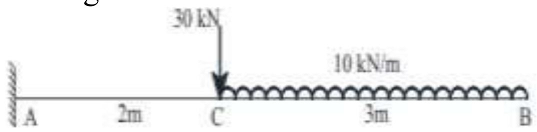


OR

	(c)	Find the position of the centroid of I-section as shown in Figure.	**07**

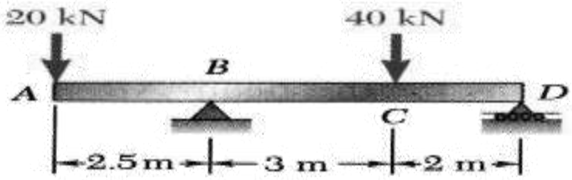


Q.3	**(a)**	Explain: (1) Types of beams (2) Types of reactions.	**03**
	(b)	State Hook's law. Draw stress strain curve for MS specimen and explain each point in detail.	**04**
	(c)	Determine the support reactions for the beam shown below. Also plot SF and BM diagrams.	**07**



OR

Q.3	**(a)**	Define stress. Also explain types of stresses.	**03**
	(b)	Determine support reaction for the given beam shown in figure below.	**04**

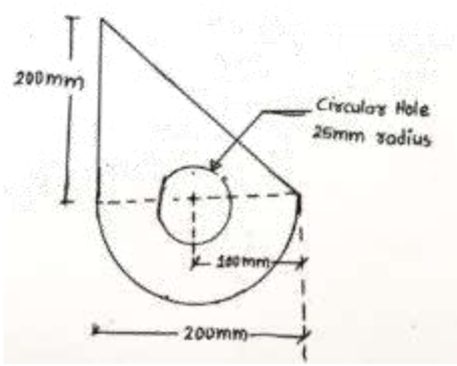


	(c)	For above figure draw SF and BM diagram with calculation.	**07**

Q.4	**(a)**	Discuss critically the assumption made in theory of Pure Bending.	**03**
	(b)	State and explain Verignon's principle.	**04**
	(c)	A reinforced concrete column is applied 700 kN load. Size of column is 250mm X 450mm, and it is reinforced with 6 bars of 20mm dia. Determine load taken by column and steel.	**07**

OR

Q.4	**(a)**	What is difference between deficient truss and Redundant truss.	**03**
	(b)	Derive the formula for the elongation of a rectangular bar under the action of axial load.	**04**
	(c)	Determine the centroid of the section shown in Figure below.	**07**



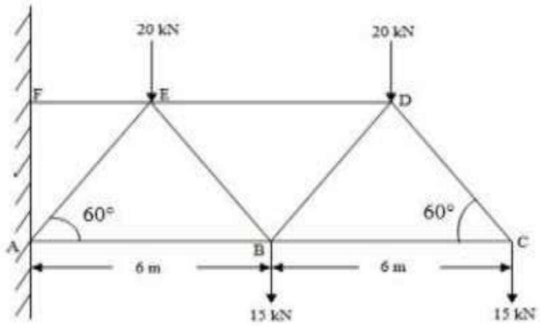
Q.5	**(a)**	State parallel axes and perpendicular axes theorems.	**03**
	(b)	Derive torsion equation with usual notations.	**04**
	(c)	Draw the mohr's stress circle for direct stresses of 70 MN/m^2 (tensile) and 40 MN/m^2 (compressive) and estimate the magnitude and direction of the resultant stresses and planes making angles of 30° and 70° with the plane of the first principal stress. Find also the normal and tangential stresses on these planes.	**07**

OR

Q.5 **(a)** Describe the Mohr's circle method to calculate principal stresses. **03**

(b) Derive assumption made in analysis of truss. **04**

(c) Determine the forces in the members DE, BE and AB of the truss, shown in figure below. **07**



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Seat No.: ________ Enrolment No.___________

GUJARAT TECHNOLOGICAL UNIVERSITY

BE - SEMESTER– III(NEW) EXAMINATION – WINTER 2022

Subject Code:3130608 Date:01-03-2023

Subject Name:Mechanics of Solids

Time:02:30 PM TO 05:00 PM Total Marks:70

Instructions:

1. **Attempt all questions.**
2. **Make suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**
4. **Simple and non-programmable scientific calculators are allowed.**

Q.1	**(a)**	Explain following terms (i) Rigid body (ii) Deformable body (iii) Elastic body.	**03**
	(b)	State and explain parallelogram law of forces.	**04**
	(c)	Determine the location of centroid of plane lamina shown in **Figure 1** with respect to point A.	**07**
Q.2	**(a)**	Write the assumptions made in the analysis of perfect truss.	**03**
	(b)	Find magnitude and direction of resultant for forces system as shown in **Figure 2.**	**04**
	(c)	Replace the forces acting on the road by an equivalent single resultant force and couple system acting at point A for **Figure 3.**	**07**
		OR	
	(c)	Find the I_{xx} and I_{yy} for section shown in **Figure 4**.	**07**
Q.3	**(a)**	Explain: (i) Type of beams (ii) Type of loading on the beams.	**03**
	(b)	Derive using first principle the equation for calculation of maximum shear stress at a section for a beam with rectangular cross section.	**04**
	(c)	Find support reaction and draw **S.F.D** and **B.M.D** for beam which is shown in **Figure 5.**	**07**
		OR	
Q.3	**(a)**	Explain following terms: (i) Shear force (ii) Bending moment (iii) Point of contra flexure.	**03**
	(b)	Find support reaction for beam which is shown in **Figure 5.**	**04**
	(c)	Find support reaction and draw **S.F.D** and **B.M.D** for beam which is shown in **Figure 6.**	**07**
Q.4	**(a)**	Explain assumptions made in theory of pure bending.	**03**
	(b)	Draw shear stress distribution diagram for Rectangular, Circular T section and I section.	**04**
	(c)	Determine deformation in each part of the bar ABCD shown in **Figure 7.**	**07**
		OR	
Q.4	**(a)**	Define stress, strain and poisson ratio.	**03**
	(b)	A solid circular shaft of 300 mm diameter has transmit 600 kW power at 200 R.P.M. Calculate maximum shear stress in shaft material.	**04**

(c) A beam of T shaped cross section is shown in **Figure 8.** is subjected to bending moment of 20 kN.m. Find the bending stress at the top and bottom of beam. **07**

Q.5 **(a)** Derive with usual notations the theorem of perpendicular axis. **03**

(b) Determine torque transmitted by hollow circular shaft of 100 mm external diameter and 70 mm internal diameter if maximum shear stress is not exceed 80 N/mm^2. **04**

(c) A beam of T shaped cross section is shown in **Figure 8.** is subjected to shear force of 50 kN. Find the maximum shear stress in section. **07**

OR

Q.5 **(a)** Define principal planes and principal stresses. **03**

(b) A steel bar of rectangular cross-section 25 mm x 40 mm carries an axial tension of 40 kN. Determine the average tensile stress in bar. **04**

(c) Determine normal and tangential stress on plane AB, in a strained material shown in **Figure 9.** **07**

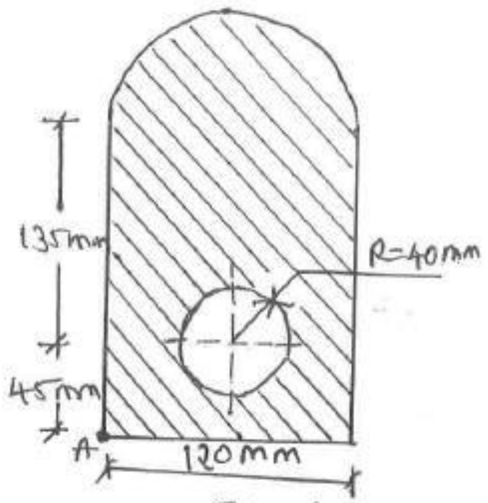

Fig-1

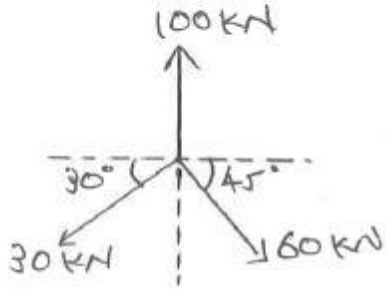

Fig-2

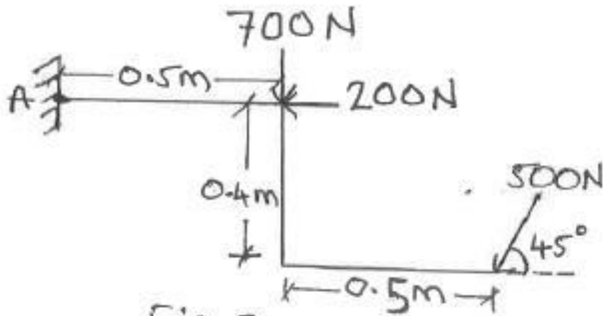

Fig-3

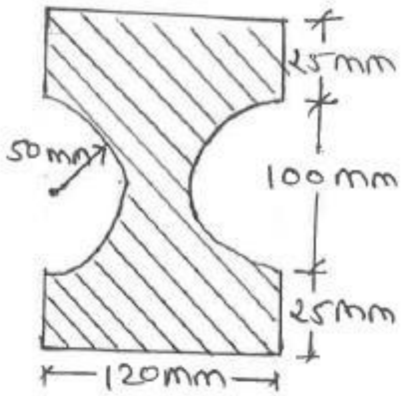

Fig-4

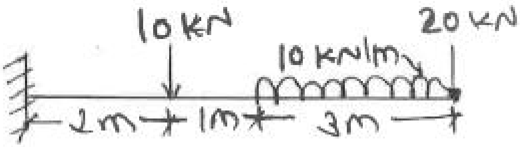

Fig-5

Fig-6

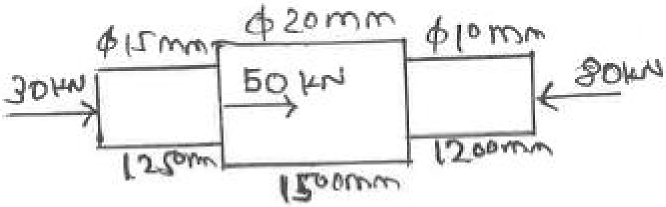

Fig. 7

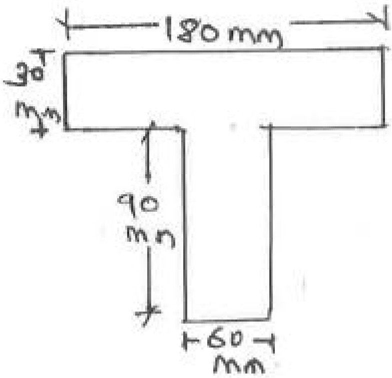

Fig-8

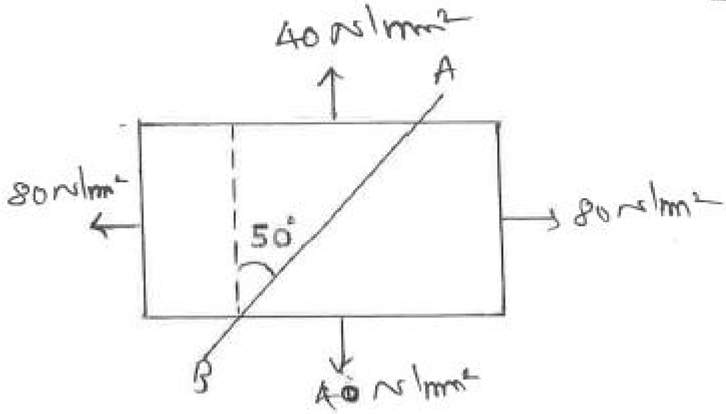

Fig-9

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Seat No.: ________ Enrolment No.___________

GUJARAT TECHNOLOGICAL UNIVERSITY

BE - SEMESTER–III(NEW) EXAMINATION – SUMMER 2023

Subject Code:3130608 **Date:03-08-2023**

Subject Name:Mechanics of Solids

Time:02:30 PM TO 05:00 PM **Total Marks:70**

Instructions:

1. **Attempt all questions.**
2. **Make suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**
4. **Simple and non-programmable scientific calculators are allowed.**

			Marks
Q.1	**(a)**	Explain free body diagram with neat sketch.	**03**
	(b)	Write various systems of forces and explain coplanar concurrent force system.	**04**
	(c)	Determine magnitude and direction of resultant force of the force system shown in **Fig.-1.**	**07**
Q.2	**(a)**	Write the assumption made in analysis of truss.	**03**
	(b)	Define shear force and bending moment with sign conventions.	**04**
	(c)	Determine magnitude, direction and position of resultant force of the force system given in **Fig.-2** with reference to point A.	**07**
		OR	
	(c)	Draw shear force and bending moment diagram for the beam shown in **Fig.-3.**	**07**
Q.3	**(a)**	Differentiate in-between centre of gravity & centroid.	**03**
	(b)	Write equation of moment of inertia for rectangular section and triangular section about its neutral axis and base of section.	**04**
	(c)	Determine the centroid of the plane area shown in **Fig.-4.**	**07**
		OR	
Q.3	**(a)**	Write assumption made in the theory of torsion.	**03**
	(b)	Find out radius of gyration for square section. Consider side dimension is **'B'** mm.	**04**
	(c)	Find the moment of inertia about both centroidal axes of Z section as shown in **Fig.-5.**	**07**
Q.4	**(a)**	Write assumptions made in theory of pure bending.	**03**
	(b)	Draw shear stress distribution diagram for hollow rectangular, hollow circle and H section.	**04**
	(c)	Calculate the diameter of the shaft required to transmit 45 kW at 120 rpm. The maximum torque is likely to exceed the mean by 30% for a maximum permissible shear stress of 55 N/mm^2. Calculate also the angle of twist for a length of 2 m. $G = 80x\ 10^3\ N/mm^2$.	**07**
		OR	
Q.4	**(a)**	Define composite beam and give main objectives of it.	**03**
	(b)	A simply supported beam 300 mm x 600 mm of 6 m. span is subjected to UDL of 15 kN/m throughout the span. Find the maximum bending stress in the beam.	**04**
	(c)	Determine the shear stress at the junction of the flange & web of an **'I'** section as shown in **Fig.-6.** Consider shear force 20 kN.	**07**
Q.5	**(a)**	Define with sketch (i) tensile stress (ii) compressive stress (iii) shear stress	**03**

	(b)	State Hook's low. Explain stress-strain behavior of mild steel with sketch.	**04**
	(c)	Determine the magnitudes & directions of principal stresses for two dimensional body as shown in **Fig.-7.**	**07**

OR

Q.5	**(a)**	Explain deformation of uniform bar section under self weight.	**03**
	(b)	Determine change in volume of a steel bar of 100 mm dia. and 500 mm length, when it is subjected to axial pull of 50 kN. Take Es = 200 GPa. & poission ratio 0.25.	**04**
	(c)	Calculate change in volume of a rectangular block 525 mm x 230 mm x 115 mm is subjected to load as shown in **Fig.-8.** Consider poission ratio 0.25 and E is 2×10^5 N/mm^2.	**07**

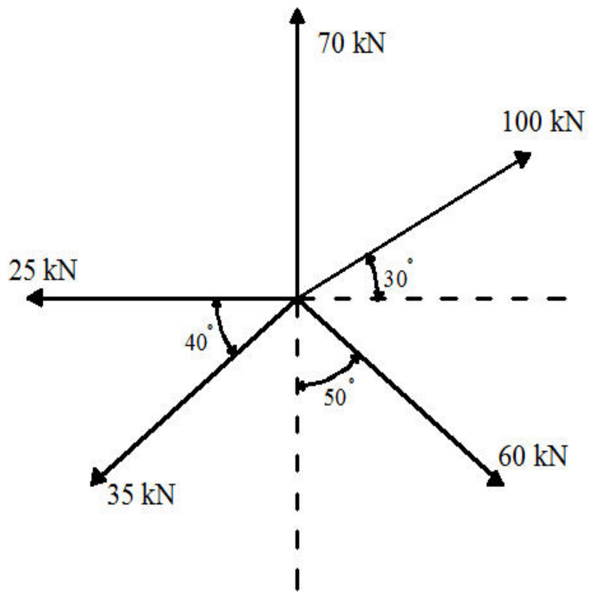

Fig. - 1

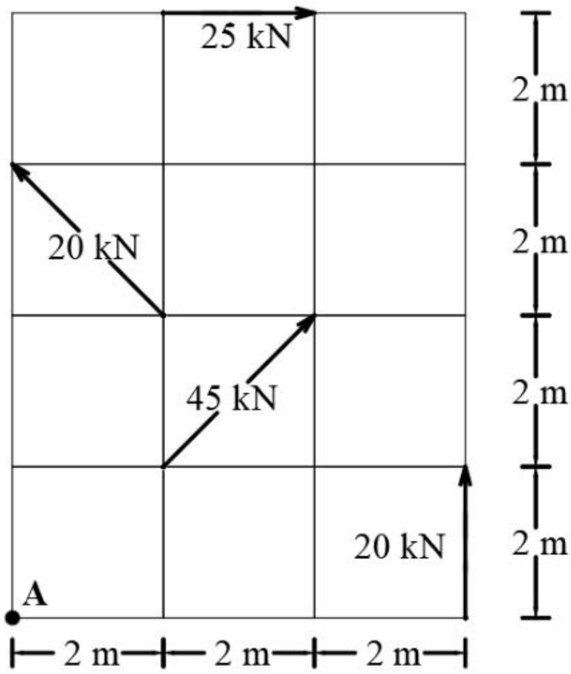

Fig. - 2

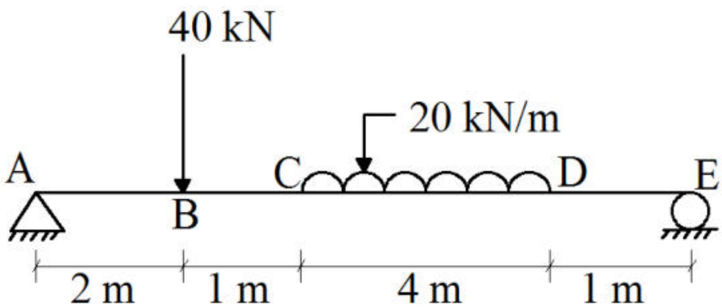

Fig. - 3

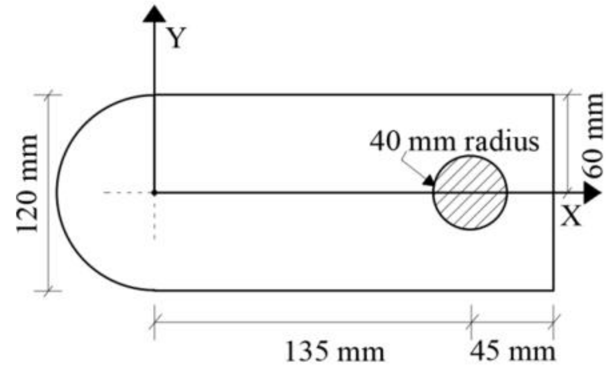

Fig. - 4

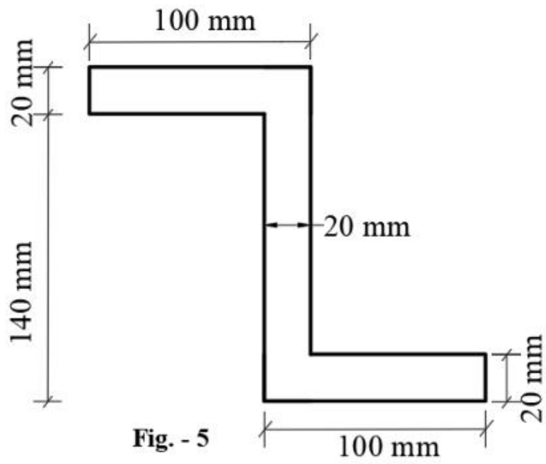

Fig. - 5

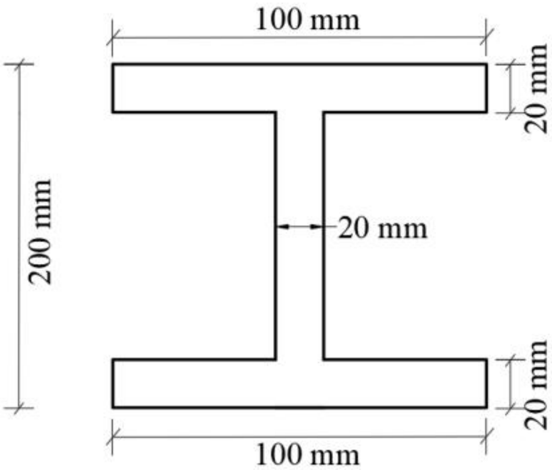

Fig. - 6

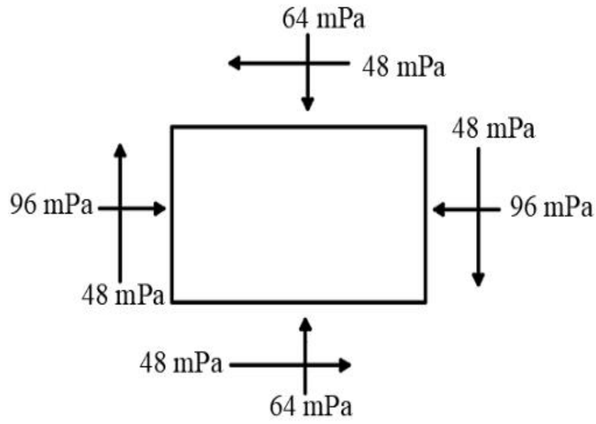

Fig. - 7

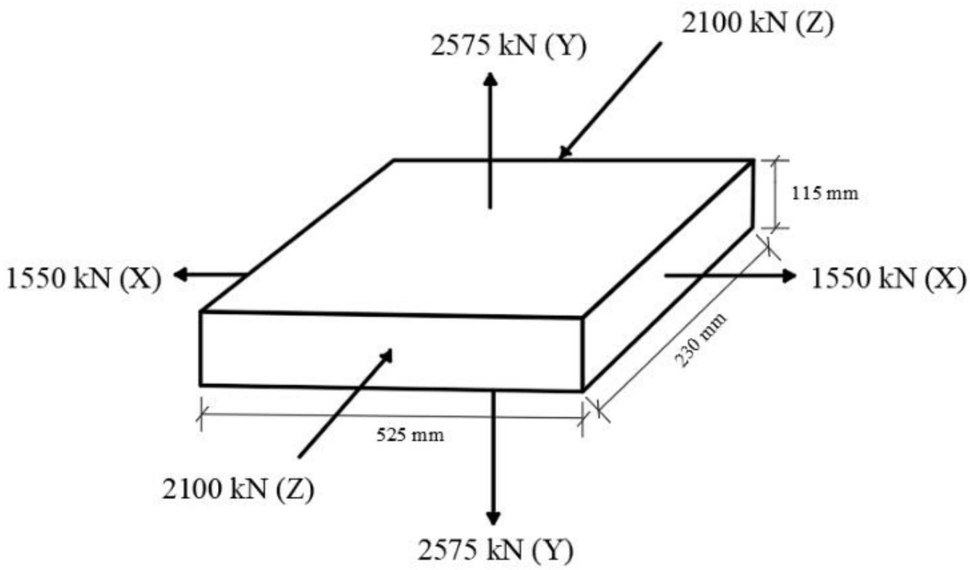

Fig. - 8

Seat No.: ________ Enrolment No.___________

GUJARAT TECHNOLOGICAL UNIVERSITY

BE - SEMESTER–III (NEW) EXAMINATION – WINTER 2023

Subject Code:3130608 **Date:23-01-2024**

Subject Name:Mechanics of Solids

Time:10:30 AM TO 01:00 PM **Total Marks:70**

Instructions:

1. **Attempt all questions.**
2. **Make suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**
4. **Simple and non-programmable scientific calculators are allowed.**

Marks

Q.1 **(a)** State following laws: **03**
1) Principle of Superposition
2) Principle of Transmissibility of Forces

(b) State and explain parallelogram law of forces. **04**

(c) For the concurrent force system shown in **Figure-1**, calculate the magnitude and direction of Resultant force **using Graphical method**. **07**

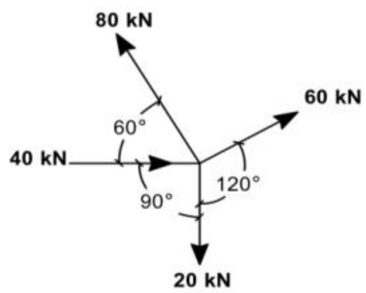

Figure-1

Q.2 **(a)** Write down assumptions made in analysis of plane trusses. Also, enlist methods of analysis of plane trusses. **03**

(b) Calculate support reactions for the beam shown in **Figure-2**. **04**

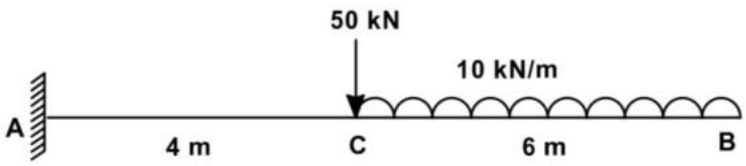

Figure-2

(c) For the beam shown in **Figure-3**, calculate support reactions and plot shear force diagram. Also, obtain value of maximum positive shear force. **07**

OR

(c) For the beam shown in **Figure-3**, calculate support reactions and plot Bending Moment Diagram. Also, obtain value of maximum bending moment in the beam. **07**

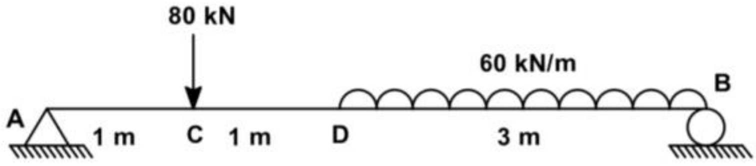

Figure-3

Q.3 **(a)** Plot shear stress distribution across the section for **03**
(1) Circular cross section (2) Symmetric 'I'-Section

(b) Obtain value of maximum shear stress at a section for rectangular cross section using first principle. **04**

(c) For the beam shown in **Figure-2**, calculate the maximum Bending stress induced at a section. Take beam cross section 300mm (width) x 600mm (depth). Also, plot bending stress variation across the section. **07**

OR

Q.3 **(a)** Write assumptions made in derivation of equation of bending. **03**

(b) Derive theory of pure bending with usual notations. **04**

(c) For the beam shown in **Figure-2**, calculate the maximum shear stress induced at a section. Take beam cross section 300mm (width) x 600mm (depth). Also, plot shear stress variation across the section. **07**

Q.4 **(a)** Give statement of Pappus-Guldinus **First theorem** and state its application. **03**

(b) Using first principle, obtain the location of centroid of semi-circular plate from the straight face of the section. **04**

(c) For the beam section shown in **Figure-4**, obtain the moment of inertia about x-x axis passing through the base of the section. **07**

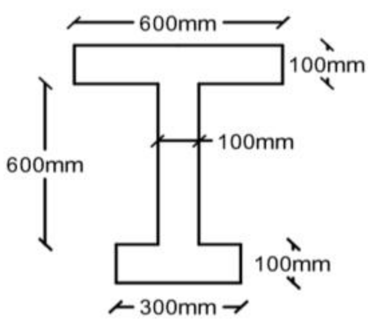

Figure-4

OR

Q.4 **(a)** State assumptions made in derivation of Torsion formula. **03**

(b) State and explain Parallel axes theorem. **04**

(c) For the beam section shown in **Figure-4**, obtain the location of centroid. **07**

Q.5 (a) Define Elasticity and State Hook's law. **03**

(b) For a two dimensional system subjected to tensile stresses along on both perpendicular directions, derive an equation to obtain principal stress on any one principal plane. **04**

(c) A steel rod having 30 mm diameter and 500 mm length is subjected to rise in temperature through 80˚C. Calculate the extension of the rod due to rise in temperature. Also, calculate the value of 'P', if the rod is held in position by axial force 'P' to prevent the expansion. Take E_s = 200GN/m^2 and α_s = 12 x 10^{-6} per °C. **07**

OR

Q.5 (a) Describe the Mohr's circle method to obtain stress at a point on an incline plane for the two dimensional system subjected to tensile stresses on both perpendicular directions. **03**

(b) For a two dimensional system subjected to tensile stresses along on both perpendicular sides, derive an equation to obtain inclination of principal plane. **04**

(c) A bar ABCD of circular section is rigidly fixed at ends A and D. The bar is subjected to axial loads as shown in **Figure-5**. Length AB = BC = CD = 400 mm. The diameter of portions AB, BC and CD are 25mm, 50mm and 75mm, respectively. Find the loads shared by each portion of the bar and change in length of each portion of the bar, neglecting any bending effects. Take E=200kN/mm^2. **07**

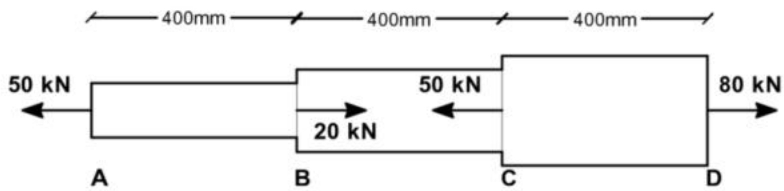

Figure-5

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Enrolment No./Seat No_______________

GUJARAT TECHNOLOGICAL UNIVERSITY

BE - SEMESTER–III (NEW) EXAMINATION – SUMMER 2024

Subject Code:3130608 **Date:04-07-2024**

Subject Name: Mechanics of Solids

Time:10:30 AM TO 01:00 PM **Total Marks:70**

Instructions:

1. **Attempt all questions.**
2. **Make suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**
4. **Simple and non-programmable scientific calculators are allowed.**

			Marks
Q.1	**(a)**	Write the assumptions made in the analysis of plane trusses.	**03**
	(b)	State following Principles (i) Principle of superposition (ii) Principle of Transmissibility	**04**
	(c)	Calculate the magnitude and direction of the resultant of a concurrent force system shown in **Figure-1**. Use analytical method.	**07**

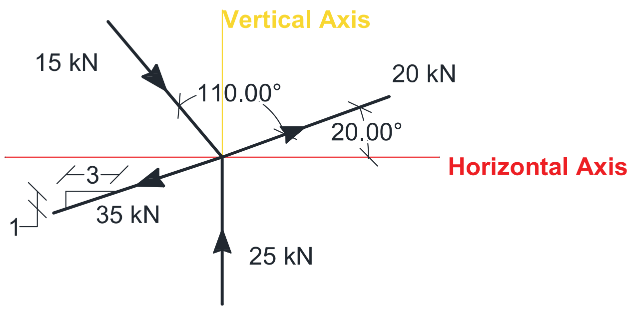

Figure-1

Q.2	**(a)**	Write down the characteristics of a couple.	**03**
	(b)	Calculate the Bending Stress at a section 2m from the left support for the beam shown in **Figure-2**. Beam cross section is 300 mm × 600 mm deep and the modulus of elasticity is 25000 MPa.	**04**
	(c)	Analyze the beam shown in **Figure-2**. Plot Bending Moment Diagram. Also, calculate maximum Bending Moment in the beam.	**07**

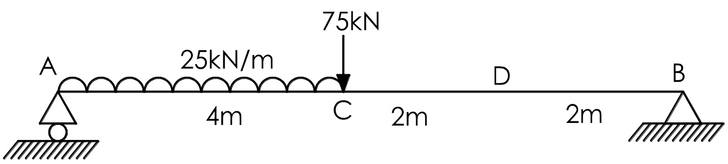

Figure-2

OR

	(c)	Analyze the beam shown in **Figure-2**. Plot Shear Force Diagram. Also, locate the point of maximum Bending Moment.	**07**
Q.3	**(a)**	Write assumptions made in the derivation of torsion equation.	**03**
	(b)	Using first principle, obtain the distance of centroid of a right-angled triangular lamina from the base.	**04**
	(c)	With usual notations, derive the formula for calculating Shear stress at a section in beams. Also, draw qualitative shear stress distribution across the section for an I-Section.	**07**

OR

Q.3 **(a)** State the Pappus-Guldinus Theorem-1 and Theorem-2. **03**

(b) Derive the torsion equation with usual notations for the circular solid shaft subjected to pure torsion. **04**

(c) Calculate the Shear Stress at a section 2m from the left support for the beam shown in **Figure-2**. Beam cross section is 300 mm × 600 mm deep and the modulus of elasticity is 25000 MPa. **07**

Q.4 **(a)** State following theorems **03**

(i) Parallel axes theorem

(ii) Perpendicular axes theorem

(b) For the two-dimensional system, a section is subjected to direct tensile stresses of σ_x and σ_y along two perpendicular directions. Derive the equation to obtain principal stresses. **04**

(c) Locate the centroid of the cross section with semicircular hole shown in **Figure-3**. **07**

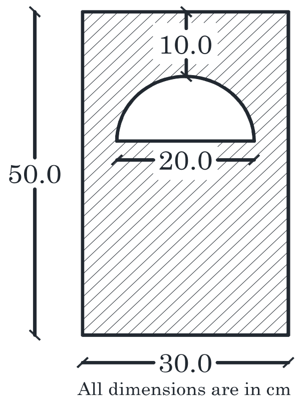

Figure-3

OR

Q.4 **(a)** Derive the equation to obtain moment of inertia of rectangular section about stronger axis of bending. **03**

(b) For the two-dimensional system, a section is subjected to direct tensile stresses of σ_x and σ_x along two perpendicular directions. Explain the step by step process of Mohr's circle method to obtain direct and tangential stresses on an inclined plane. **04**

(c) Obtain moment of inertia about centroidal x-axis for the section shown in **Figure-3**. **07**

Q.5 **(a)** Derive the relation among modulus of elasticity, modulus of rigidity and Poisson's ratio with usual notations. **03**

(b) A wire is tied straight between two rigid poles 10m apart has initial tensile stress $10 N/mm^2$ at 32° C. Calculate stress in wire if temperature reduces to 27° C. Take modulus of elasticity $(E)=2.1\times10^5 N/mm^2$ and the coefficient of thermal expansion $(\alpha)=20\times10^{-6}/°C$. **04**

(c) A point in two-dimensional stressed body is shown in **Figure-4**. Determine the magnitudes and directions of principal stresses, using analytical method. **07**

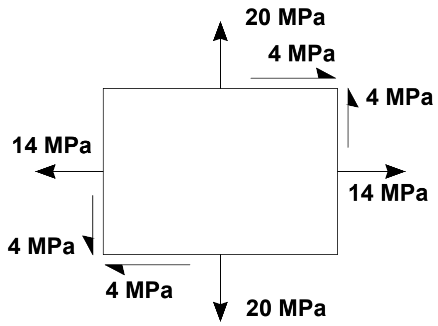

Figure-4

OR

Q.5 **(a)** A copper rod of 150mm diameter and 5m length is subjected to an axial pull of 500kN. Calculate stress and change in length of bar, if the modulus of elasticity $E=100000 N/mm^2$. **03**

(b) Derive the relation among modulus of elasticity, Bulk Modulus and Poisson's ratio with usual notations. **04**

(c) A member is formed by connecting end to end a 200mm long steel bar of 50mm × 50mm square section with 500 mm long aluminum bar of 100mm × 100mm square section as shown in **Figure-5**. Determine the total change in length of the member, if the member carries an axial tensile load of 1000kN. Also calculate stress in each part of the member. Take $E_{Steel} = 2\times10^5$ MPa and $E_{Aluminum}=1\times10^5$ MPa. **07**

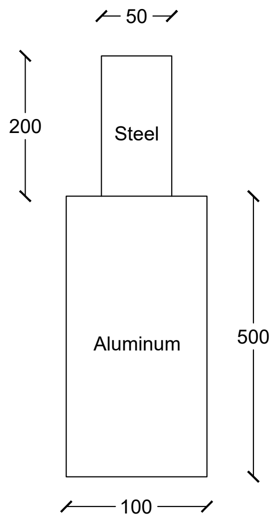

Figure-5

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Enrolment No./Seat No_______________

GUJARAT TECHNOLOGICAL UNIVERSITY

BE- SEMESTER–III (NEW) EXAMINATION – WINTER 2024

Subject Code: 3130608 **Date: 06-12-2024**

Subject Name: Mechanics of Solids

Time: 10:30 AM TO 01:00 PM **Total Marks: 70**

Instructions:

1. **Attempt all questions.**
2. **Make suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**
4. **Simple and non-programmable scientific calculators are allowed.**

			MARKS
Q.1	**(a)**	Define following terms; i) Rigid body ii) Mass iii) Equilibrant	**03**
	(b)	State any two assumptions made in analysis of plane trusses. Plot neat sketch of perfect truss, deficient truss and redundant truss.	**04**
	(c)	Obtain the magnitude and direction of equilibrant force for the force system shown in Figure-1.	**07**

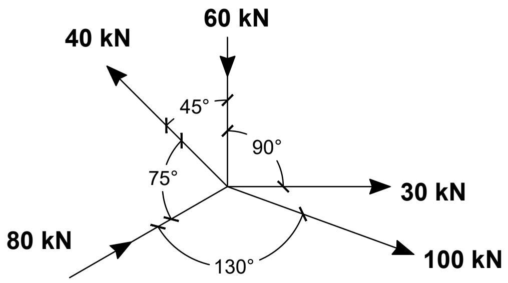

Figure-1

Q.2	**(a)**	State and explain the Law of polygon of forces.	**03**
	(b)	Derive the relation, with usual notations, among loading, shear force and bending moment at a section for a determinate beam.	**04**
	(c)	Calculate support reactions for the beam shown in Figure-2. Also plot Bending Moment and Shear Force diagrams showing values at important locations.	**07**
		OR	
	(c)	Calculate support reactions for the beam shown in Figure-2. Also obtain the location of maximum Bending Moment, Maximum Bending Moment and Maximum Shear Force in the beam.	**07**

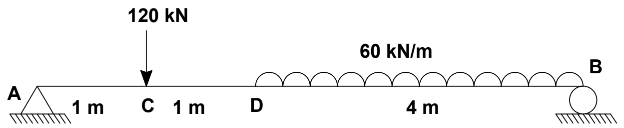

Figure-2

Q.3	**(a)**	State Pappus-Guldinus Theorems.	**03**
	(b)	From first principle obtain the location of center of gravity of semi-circular lamina.	**04**
	(c)	Calculate the bending stress at point ‘C’ for the beam shown in Figure-2. Given that the beam cross section 300mm wide × 600mm deep and modulus of elasticity as 25000MPa.	**07**

OR

Q.3 **(a)** State parallel axes theorem and perpendicular axes theorem. **03**

(b) Using first principle obtain the equation to obtain moment of inertia of a circular section about centroidal x-axis. **04**

(c) Calculate the shear stress at point 'D' for the beam shown in Figure-2. Given that the beam cross section 300mm wide × 600mm deep and modulus of elasticity as 25000MPa. **07**

Q.4 **(a)** Define the following terms; **03**

i) Modulus of elasticity
ii) Poisson's ratio
iii) Modulus of rigidity

(b) With usual notations, derive the relationship among Modulus of elasticity, Poisson's ratio and Bulk modulus. **04**

(c) A solid shaft of 250 mm diameter has the same cross-sectional area as that of hollow shaft of the same material of inside diameter 200 mm. Find the ratio of power transmitted by the two shafts at same angular velocity. Also, compare angle of twist in equal lengths of these shafts when stressed to the same intensity. **07**

OR

Q.4 **(a)** Give assumptions made in the theory of torsion. **03**

(b) For the shaft subjected to pure torsion, derive the torsion equation with usual notations. **04**

(c) Calculate the stress and elongation of each part of the bar shown in Figure-3. Also, calculate total change in length of the bar. Take E = 200000MPa. **07**

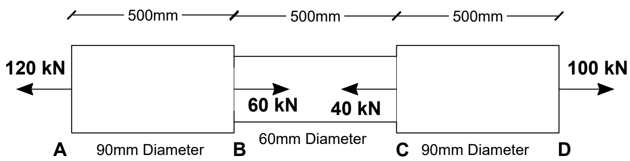

Figure-3

Q.5 **(a)** State and explain Hook's law. **03**

(b) Explain Mohr's circle method with neat sketch for a section subjected to direct stresses along two perpendicular directions. **04**

(c) A bar held straight between two rigid supports 5 m apart has initial tensile stress 5 MPa at 30° Celsius. Calculate the stress in wire if temperature reduces to minus 5° C. Take E= 2×10^5 MPa and coefficient of thermal expansion as 2×10^{-6} per °C. **07**

OR

Q.5 **(a)** Define Stress, Strain and Bulk modulus. **03**

(b) Explain Mohr's circle method with neat sketch for a section subjected to direct stresses along two perpendicular directions and accompanying shear stresses. **04**

(c) Determine the compressive stress developed in a punch of 10 mm diameter, used to make a hole of 10mm diameter in 6mm thick mild steel plate. The shear strength of mild steel is 200MPa. **07**

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Enrolment No./Seat No_______________

GUJARAT TECHNOLOGICAL UNIVERSITY

BE - SEMESTER–III EXAMINATION – SUMMER 2025

Subject Code:3130608 **Date:04-06-2025**

Subject Name:Mechanics of Solids

Time:02:30 PM TO 05:00 PM **Total Marks:70**

Instructions:

1. **Attempt all questions.**
2. **Make suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**
4. **Simple and non-programmable scientific calculators are allowed.**

			MARKS
Q.1	**(a)**	Define : (i) Space (ii) Resultant (iii) Couple	**03**
	(b)	Determine resultant of coplanar concurrent force system shown in fig.1	**04**
	(c)	Find support reactions for the beam shown in fig.2	**07**
Q.2	**(a)**	Enlist Fundamental principles of mechanics. State principle of transmissibility.	**03**
	(b)	A cord supported at A and B carries a load of 10 kN at D and a load of W at C as shown in Fig.3. Find the value of W so that CD remains horizontal.	**04**
	(c)	Analyse the truss shown in fig. 4	**07**
		OR	
	(c)	For the system of force on a lamina OABC shown in figure 5, find magnitude and direction of the resultant force. Also locate the resultant by perpendicular distance from point "O".	**07**
Q.3	**(a)**	Determine resultant of coplanar concurrent force system shown in fig.1 using graphical method.	**03**
	(b)	Derive relation between uniformly distributed load, shear force and bending moment with usual notations.	**04**
	(c)	Draw shear force and bending moment diagrams for the beam shown in fig. 6	**07**
		OR	
Q.3	**(a)**	Define (i) Shear force (ii) Point of zero shear (iii) Point of contraflexure	**03**
	(b)	State assumption made in theory of pure bending.	**04**
	(c)	Find centroid of lamina shown in fig. 7	**07**
Q.4	**(a)**	State and explain Pappus Guldinus first theorem using appropriate example.	**03**
	(b)	A rectangular beam 300 mm deep is simply supported over a span of 4.0 m. What uniformly distributed load the beam may carry if the bending stress is not to exceed 120 MPa? Take I = 8 x 10^6 mm^4	**04**
	(c)	Calculate stresses in each portion and the total change in length for steel bar ABCD as shown in figure 8. Take E = 200 GPa	**07**
		OR	
Q.4	**(a)**	Define : (i) Shear Stress (ii) Modulus of Rigidity (iii) Volumetric strain	**03**
	(b)	Determine moment of inertia about its horizontal centroidal axis for T section having flange and web dimensions 100mm x20 mm each.	**04**
	(c)	A short concrete column 300mm x 300mm in section is carrying axial load of 360 kN. The column is reinforced by four 12mm diameter steel bars each one at corner. Calculate stresses in concrete and steel.	**07**

Take Ec = 14GPa and Es = 210Gpa.

Q.5 **(a)** Sketch qualitative shear stress distribution diagrams for following sections (i) Circular (ii) I section and (iii) T section **03**

(b) A steel tube of 2 m length is subjected to 50º C rise in temperature. Determine (i) free natural expansion and (ii) stress developed in the tube, if expansion is prevented. Take Es = 2.0 x 10^5 N/mm^2 and α = 12 x 10^{-6} per º C. **04**

(c) Define: (i) Poisson's ratio (ii) Bulk modulus (iii) Modulus of Elasticity. Derive relation between bulk modulus, Poisson's ratio and modulus of elasticity. **07**

OR

Q.5 **(a)** Define (i) Torsional Rigidity (ii) Principal Plane (iii) Neutral axis **03**

(b) A solid steel shaft of 60 mm diameter is subjected to torque of 5 kNm. Determine maximum shear stress developed in the shaft. G= 80GPa **04**

(c) For an element shown in fig.9. **07**
Determine (i) Principal stresses and location of corresponding principal planes.
(ii) Maximum shear stress and location of planes containing it

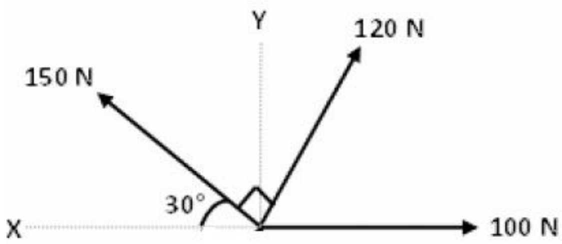

Fig. 1

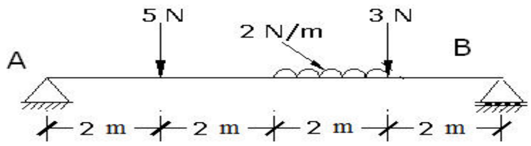

Fig. 2

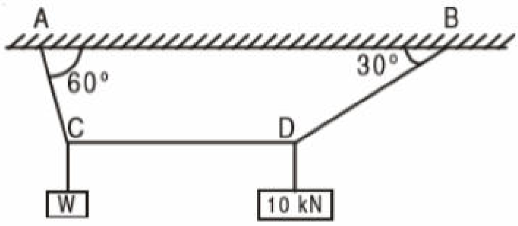

Fig. 3

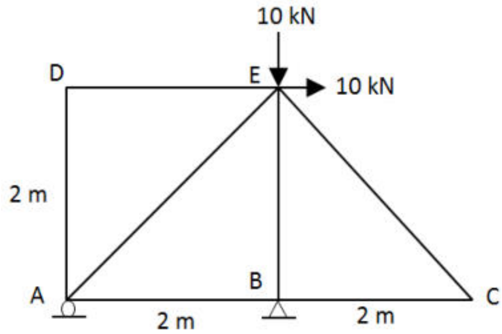

Fig. 4

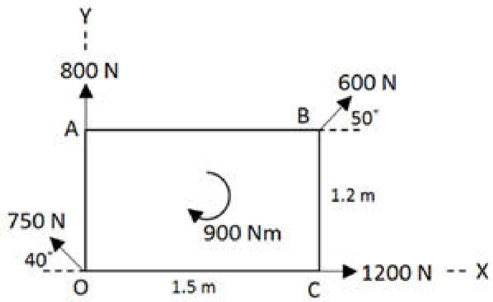

Fig. 5

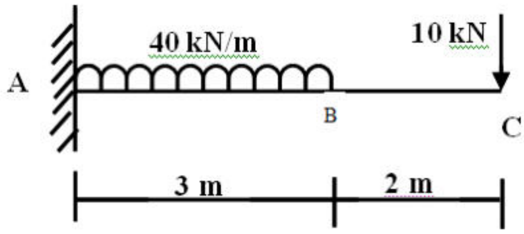

Fig.6

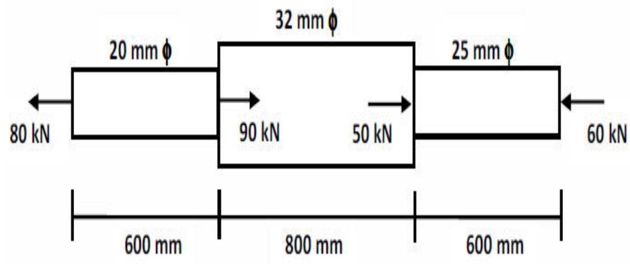

Fig. 8

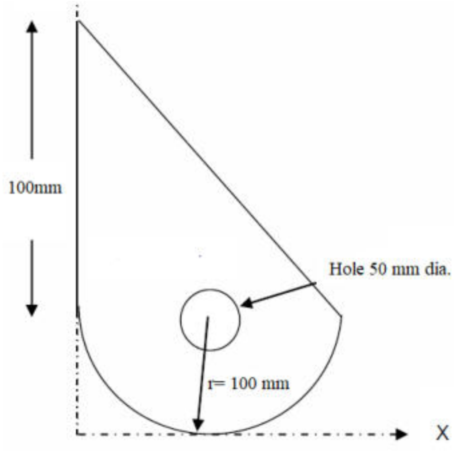

Fig. 7

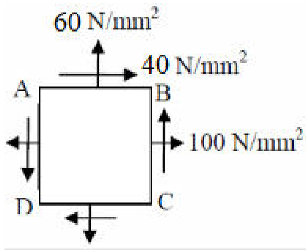

Fig. 9

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Enrolment No./Seat No_______________

GUJARAT TECHNOLOGICAL UNIVERSITY

BE- SEMESTER–III EXAMINATION – WINTER 2025

Subject Code:3130608 **Date:19-12-2025**

Subject Name: Mechanics of Solids

Time:10:30 AM TO 01:00 PM **Total Marks:70**

Instructions:

1. **Attempt all questions.**
2. **Make suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**
4. **Simple and non-programmable scientific calculators are allowed.**

Q.1 **(a)** State and explain The Lami's theorem. **03**

(b) State and explain **04**

(i) Principle of transmissibility of force,

(ii) Principle of superposition.

(c) Fill in the blanks : **07**

(i) At neutral axis , bending stress in the beam is _______

(ii) The shear taken by the web of an I section is ________ flange.

(iii) ________ Beams have one end fixed and the other free.

(iv) The value of bending moment at the point of contra flexure will always be ___

(v) A _____has the same effect as the combined effect of forces that it replaces.

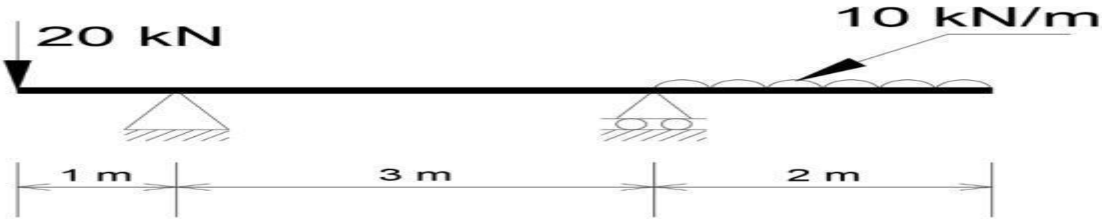
(vi) Mathematical equation of perpendicular axis theorem is ______

(vii) Frictional force will always be in a direction ___ to that in which the body tends to move.

Q.2 **(a)** Define: (1) Shear Force (2) Bending Moment (3) Points of contra flexure. **03**

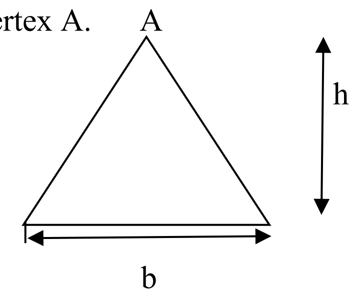
(b) Derive the relationship between rate of loading, shear force and bending moment. **04**

(c) Draw shear force diagram and bending moment diagram for a beam shown in fig.1 **07**



OR

(c) Find the moment of inertia of a triangular area about its centroidal axes and about the vertex A. **07**



b

Q.3 **(a)** State and explain with an example the Pappus-Guldinus theorems. **03**

(b) A steel bar of 1 meter length is subjected to 120 kN axial tensile force. The C/S of bar is 20 mm x 20 mm. The increase in length is found to be 0.5 mm and decrease in thickness is 0.003 mm. Find the value of Young's modulus and poisson's ratio. Thi **04**

(c) Define following : **07**

i. Moment of inertia of a section,
ii. Polar moment of inertia,
iii. Radius of gyration
iv. Section Modulus

OR

Q.3 **(a)** Using first principle, obtain the distance of centroid of a right-angled triangular lamina from the base. **03**

(b) Find the support reaction for beam which is loaded as shown in **Fig.** **04**

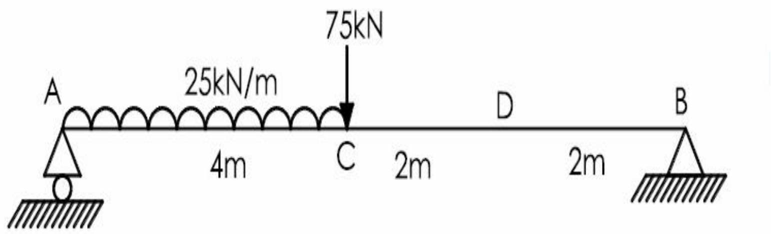


Fig. 2

(c) Calculate the maximum bending stress and maximum shear stress at a section for the beam shown in Figure-2. Beam cross section is 300mm wide × 500mm deep. **07**

Q.4 **(a)** Write down assumptions made in analysis of plane trusses. Enlist methods of analysis of plane trusses. **03**

(b) Define and explain significance of following terms/quantities. **04**

i Volumetric strain,
ii Poisson's ratio,
iii Lateral strain,
iv Modulus of elasticity,

(c) A simply supported beam of span 4.0 m having uniform section of size 200 mm width and 400 mm depth is loaded with uniformly distributed load of 40 kN/m over entire span. Determine the maximum bending moment along the span and draw the bending stress distribution across the section. **07**

OR.

Q.4 **(a)** Derive the formula for shear stress distribution in a rectangular beam section. **03**

(b) Derive theory of pure bending with usual notations. **04**

(c) A "T" section has a flange 160 mm X 12.5 mm and web 188 mm X 8 mm. It is used as a beam over span of 4.0 m to carry uniform load of 16 kN/m. Sketch the shear stress distribution at the section of maximum shear force. **07**

Q.5 **(a)** Describe the Mohr's circle method to calculate principal stresses. **03**

(b) An elastic material is subjected to two direct stresses of 200 N/ mm^2 and 80 N/ mm^2 tensile at right angles to each other. If major principal stress is limited to 210 N/ mm^2 compressive, find the value of shear stress that can be applied to the material. Also find minor principal stress. **04**

(c) Derive the torsion formula for cylindrical shaft. State the assumptions taken in derivation of torsion formula. Define the torsional rigidity. **07**

OR.

Q.5 **(a)** State and explain Varignon's principal of moments. **03**

(b) Derive the equation for deformation and stresses in composite structures. **04**

(c) A solid circular shaft has to transmit 300 kW power at 210 r.p.m. The limiting shear stress is 50 N/mm^2 and the permissible angle of twist is 1° in 3.0 m length of the shaft. Determine the minimum diameter of shaft required to transmit the power. Take G = 0.8 X 10^5 N/mm^2. **07**

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